

Verdant Search: System Demonstration

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Demonstration Agenda

1. Module 2: Search and Retrieval

- Hybrid search (BM25 + Vector)
- Search results display

2. Module 3: Data Ingestion (Partial)

- Document indexing process
- Index monitoring

3. Module 5: AI Iterative Chat (RAG)

- Context-aware answer generation
- Citation and source references
- Iterative query refinement

4. Database Demonstration

Module 2: Search and Retrieval - Implementation Status

Implemented Features:

- ✓ **Text Input & Query Processing**
 - User query submission through search interface
 - Query sanitization and normalization
 - Query expansion with synonyms
- ✓ **Keyword Search (BM25)**
 - Inverted index with term posting lists
 - BM25 probabilistic ranking
 - Fast lexical matching
- ✓ **Vector Search (HNSW)**
 - Sentence-BERT query encoding
 - HNSW approximate nearest neighbor search
 - Semantic similarity matching

Module 2: Search and Retrieval - Implementation Status (Cont.)

Implemented Features (Continued):

- ✓ **Hybrid Search Fusion**
 - Reciprocal Rank Fusion (RRF) combining BM25 and vector scores
 - Weighted score combination
 - Metadata-based pre-filtering
- ✓ **Search Results Display**
 - Ranked document list with relevance scores
 - Document snippets with highlighted keywords
 - Metadata display (filename, page number, timestamp)

Module 2: Search and Retrieval - Implementation Status (Final)

Implemented Features (Final):

- ✓ **Conversational Search**
 - Follow-up question support
 - Conversation history tracking
 - Query rewriting for context preservation
- ✓ **Metadata Filtering**
 - Filter by document type, date, source
 - Dynamic faceted search

Ready for Demonstration

All core search and retrieval features are fully implemented and operational.

Module 3: Data Ingestion - Partial Implementation Status

Implemented Features:

- ✓ **Document Chunking**
 - Semantic segmentation of documents
 - Chunk overlap for context preservation
 - Support for heading-based chunking
- ✓ **Index Building**
 - Inverted index construction for BM25 (Pyserini)
 - HNSW vector index building (FAISS)
 - Dense vector embedding generation (Sentence-BERT)
- ✓ **Index Monitoring**
 - Real-time indexing progress tracking
 - Index health and size monitoring
 - Document count statistics

Module 3: Data Ingestion - Partial Implementation Status (Cont.)

Implemented Features (Continued):

- ✓ **Scheduled Updates**
 - Cron-based index refresh scheduling
 - Batch update support

Demonstration Ready

Index building and monitoring features are operational and ready for demonstration.

Module 5: AI Iterative Chat (RAG) - Implementation Status

Implemented Features:

- ✓ **Query Processing & Validation**
 - User query reception and validation
 - Query length and format checking
- ✓ **Context Injection**
 - Retrieved context injection into LLM prompt
 - Context formatting with document metadata
 - Structured prompt engineering
- ✓ **AI Answer Generation**
 - LLM-based answer generation grounded in context
 - Hallucination prevention (context-only responses)
 - Markdown-formatted answer display

Module 5: AI Iterative Chat (RAG) - Implementation Status (Cont.)

Implemented Features (Continued):

- **✓ Citation & Source References**
 - Inline citations with source document links
 - Citations include page numbers and timestamps
 - Highlighted cited text in source documents
 - Formatted source references (title, author, timestamp)
- **✓ Real-time Streaming**
 - Server-sent events (SSE) for streaming responses
 - Progressive answer display
 - Low latency user experience
- **✓ Knowledge Base Awareness**
 - Clear indication of knowledge base limitations
 - Entailment checking for answer grounding validation

Module 5: AI Iterative Chat (RAG) - Implementation Status (Final)

Implemented Features (Final):

- **✓ Iterative Conversation**
 - Follow-up question support
 - Intent detection (follow-up vs. new query)
 - Conversation history maintenance (last 10 turns)
- **✓ Query Refinement Loop**
 - LLM-based query rewriting with conversation context
 - Automatic retrieval triggering after rewriting
 - Dynamic document list updates
- **✓ Feedback Integration**
 - User feedback analysis
 - Search strategy refinement based on feedback

Database Demo

More To-Do in FYP2

Planned Enhancements:

- **Complete Data Ingestion Module**

- Distributed crawling from multiple sources (Confluence, Google Drive, Git)
- Text and image extraction from documents
- OCR support for scanned PDFs

- **Implement Image-Supported Search + Reranking**

- CLIP-based image embeddings
- Multimodal reranking with BLIP-2
- Image-text cross-modal relevance scoring

- **Implement Chatbot Image Conversational Ability**

- Vision-language model integration for image understanding
- Image-aware context injection
- Multimodal citation with image references

Thank You!

Questions?

Verdant Search: AI Boosted Sharded Distributed Search Engine
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